Dennis **Eriksson, Professor**

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Interests

Algebraic geometry
Arithmetic geometry
Complex geometry
Arakelov theory
Intersection theory
Calabi–Yau manifolds
Hodge theory
Analytic torsion
Enumerative mirror symmetry
Singularities

About me

I am a Swedish mathematician with extensive international research experience, and a strong interest in collaboration, communication and education. Outside mathematics, I enjoy language learning and cultural travel.

Languages

Swedish ●●●●
English ●●●●
French ●●●●
German ●●○○
Chinese ●○○○
Japanese ●○○○
Polish ●○○○

Research Summary

My research grows out of arithmetic geometry, where algebraic geometry, number theory, and complex analysis meet. This area combines tools from Grothendieck's algebraic geometry with ideas from complex differential geometry, and I use this interaction to study structural invariants of algebraic varieties. In recent years, these methods have naturally connected my work to mathematical aspects of mirror symmetry — a conjectural framework originating in string theory that continues to influence several areas of modern geometry — as well as to questions about singularities. Across these topics, my work studies how algebraic geometry, and in particular intersection theory, serves as a meeting point for arithmetic ideas, complex geometric methods, Hodge theory, singularities, and mirror symmetry.

Select Significant Publications

2024 **Deligne-Riemann-Roch and intersection bundles**
J. Éc. polytech. Math. 11 (2024), 247–361 · With Gerard Freixas i Montplet.

2022 **On genus one mirror symmetry in higher dimensions and the BCOV conjectures**
Forum Math. Pi 10(e19) (2022), 1–53 · With Gerard Freixas i Montplet, Christophe Mourougane.

2022 **Complex Chern–Simons bundles in the relative setting**
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2021 **BCOV invariants of Calabi–Yau manifolds and degenerations of Hodge structures**
Duke Mathematical Journal 170(3) (2021), 379–454 · With Gerard Freixas i Montplet, Christophe Mourougane.

2021 **Spaces of norms, determinant of cohomology and Fekete points in non-Archimedean geometry**
Advances in Mathematics 378 (2021), 107501–107501 (Article ID 107501) · With Sébastien Boucksom.

Some Select Research Grants

2027–2030 **Vetenskapsrådet Research Project Grant** — PI
SEK 4,292,000 (approx. €380,000), project "Invariants and geometry in mirror symmetry at genus one".

2026 **Research in Paris** — Institut Henri Poincaré
2 month invitation.

2025 **Vergstiftelsen Postdoc Grant** — Postdoc grant
Grant, approx. €200,000, to hire a postdoc for two years, in the project "The hunt for new invariants in the intersection of arithmetic and algebraic".

2022–2025 **Vetenskapsrådet Research Project Grant** — PI
Approx. €350,000.

2009–2011 **Vetenskapsrådet Postdoc Grant** — PI
Salary for two years.

2024 **G.S. Magnussons Grant** — mobility grant
Approx. €30,000.

2025 **Svensk-Franska stifelsen** — mobility grant
Approx. €6,000.

2020 **Knut och Alice Wallenberg Foundation** — Visiting Professor Grant
For Gerard Freixas i Montplet, one year visiting professor at Chalmers.

2018 **Research in Paris** — Institut Henri Poincaré
Two-month funded visit.

2014 **Trimester Program Organizer** — HIM Bonn
Group-leader: "p-adic methods in Arakelov geometry and Shimura varieties".

Positions

2024–present **Professor (Gothenburg)** — Chalmers University of Technology
Department of Mathematical Sciences.

2018–2023 **Associate Professor (Docent) (Gothenburg)** — Chalmers University of Technology
Department of Mathematical Sciences.

2014–2018 **Assistant Professor (Gothenburg)** — Chalmers University of Technology

2017 **Visitor MPI** — Max Planck Institute for Mathematics, Bonn
January–April.

2014 **Junior Trimester Program (Organiser)** — Hausdorff Institute for Mathematics
Trimester: Algebraic geometry (p-adic methods in Arakelov geometry and Shimura varieties).

2013 **Postdoctoral Researcher** — Max Planck Institute for Mathematics
January–December.

2009–2012 **Postdoctoral Researcher** — University of Gothenburg
Teaching assistant 2011–2012.

2008–2009 **Postdoctoral Researcher** — University of Tokyo

2007–2008 **JSPS Research Fellow** — University of Tokyo

Education

2003–2008 **PhD in Mathematics** — Université d'Orsay, Paris 11, Paris
Thesis: Formule de Lefschetz fonctorielle et applications géométriques, under the supervision of Damian Rössler.

2000–2003 **Undergraduate and master studies in Mathematics**
University of Gothenburg, Gothenburg + Paris 13, Paris
Thesis: Some more or less explicit class field theory. Under the supervision of Peter Hegarty.

Academic & Organizational Responsibilities

2018–2023 **PhD Student Admissions Committee** — Chalmers
Member of the PhD committee deciding future PhD students.

2019–2023 **Programme Board Member** — Chalmers
Member of the "Programråd" for Samhällsbyggnadsteknik.

2014 **Group Leader (Junior Trimester Program)** — Hausdorff Institute for Mathematics
Organized and led the Junior Trimester Program: p-adic methods in Arakelov geometry and Shimura varieties.

2009–2020 **High School Outreach Coordinator** — Gothenburg region
Contact person for high school research projects in mathematics.

2014– **Main Organizer / Co-Organizer (International Workshops)** — International workshops
N-cube (2015, 2016, 2018, 2025); Freixit 2024; Number theory and dynamical systems (2016); Algebraic Geometry (Bonn, 2018); Workshop: Arithmetic intersection theory and Shimura varieties (2014).

2010– **Thesis Committees & Examination** — Paris and Gothenburg
Rapporteur and Member of jury for Andres Gomez (Aarhus 2026); Member of jury for Juan Felipe Castro Cárdenas (Paris 2026); Rapporteur for Dai Imaike (Kyoto 2024); Rapporteur and Member of jury for Dorian Ni (Paris 2022); Member of jury for Bertrand Duma (Paris 2012); Diskutant/Opponent at licentiate defense for Ida Säfström (Gothenburg 2010).

2010– **Professional Service and Journal Refereeing** International Journals

Regular ad-hoc referee for journals including: Inventiones Mathematicae, IMRN, Algebra & Number Theory, Journal of Differential Geometry, JEMS, JHEP, JGP, Selecta Mathematica, and Pacific Journal of Mathematics. Also served as a referee for various international conference proceedings.

Teaching and Pedagogical Qualifications

2016– **Undergraduate** Chalmers

Preparatory Calculus; Calculus for Civil Engineering; Multivariable Calculus.

2016– **Master level** Chalmers, Hangzhou, Paris

Algebraic Geometry; reading courses in algebraic topology and homological algebra.

2016– **PhD level** Graduate school

K-theory (with M. Goffeng); p-adic methods (with A. Södergren, S. Herrero); reading courses in algebraic number theory, Arakelov theory, Hodge theory, derived geometry, and Gromov–Witten theory.

2019, 2020, 2023 **Teaching Award, three times** Chalmers

Prize for best teacher in the Civil Engineering programme. Nominated for the Chalmers pedagogical prize in 2020.

2024–present **Bachelor Thesis Examiner** Chalmers & University of Gothenburg

Examiner for bachelor theses in mathematics; approximately 80 students and 20 projects every year.

2018 **Pedagogical Research Publication** Primus 28 (2018), no. 8

Communicating to Learn Multivariable Calculus: Students' Blackboard Presentations as a Means for Enhancing Mathematics Learning. With Hans Malmström.

2018 **Innovation Grant** Chalmers

Approx. €70,000 for developing digital teaching tools. I applied for and led the project.

2014–2018 **Pedagogical Studies in Higher Education (15 hp)** Chalmers

Formal qualification in university-level teaching and learning, including student assessment, curriculum design, and learning theories.

Supervision

Postdocs Yanbo Fang (Current, 2026–present)

PhD students Tudor Sharpe (Current, 2026–present); Gabriel Abánades Joglar (Current, 2025–present); Anthony Mäkelä (Current, 2025–present, Co-supervisor); Douglas Molin (Completed 2026, Co-supervisor); Mykola Pochekai (Completed 2024); Jiacheng Xia (Completed 2021, Co-supervisor); Manh Hung Tran (Completed 2020)

Master students Samuel Plumeyer (Current); Sebastian Jansson (Current); Liu Xudong (2024); Felix Frelot (2022); Robert Antonio Pop Gorea (2021); Magnus Fries (2020); Mattias Lindby (2016)

Bachelor projects Louis Dao, Anna Davoodi, Erik Samuelsson (2024)

Recent Selected Seminars

Planned	**Mini-course (invited)**	Xiamen, China
Oct 2026	**How many rational points? The work of Abel prize winner Gerd Faltings** Colloquium	Gothenburg, Sweden
Sep 2026	**Genus one mirror symmetry for toric varieties**	IBS, Pohang, South Korea
Dec 2025	**Algebraic geometry seminar** On smooth fillings of Calabi–Yau varieties	Bologna, Italy
March 2025	**Séminaire de géométrie "Anniversaire Paul 80"** The spectral genus and an inequality for the Milnor number	Paris, France
March 2024	**Spring school in Arakelov Geometry and related topics** Relative intersection theory and Arakelov geometry, minicourse	Hangzhou, China
April 2024	**Algebraic geometry seminar** From counting genus one curves to questions about singularities	Stockholm, Sweden
Sep 2023	**Masterclass: Arakelov geometry on Shimura varieties** Eisenstein series and enumerative geometry	Copenhagen, Denmark
2019–2021	**Genus one mirror symmetry and the arithmetic Riemann–Roch theorem**	Princeton, Marseille, Oxford, Stockholm

Selected Conferences

Late 2027	**Conference in honour of Ken-Ichi Yoshikawa's 60th birthday** Invited	Kyoto, Japan
Sep 2026	**Arakelov geometry intercity conference** Archimedean heights and bad reduction of Calabi–Yau manifolds	Westlake University, Hangzhou, China
May 2026	**AI for Mathematics** The very confident intern — how I use LLMs in my mathematical research	Gothenburg, Sweden
Sep 2025	**Kähler Geometry: Past, Present and Future** On a conjecture of Deligne for the determinant of the cohomology	Banff, Canada
May 2025	**Hodge theory in a broad sense** On smooth fillings of Calabi–Yau families	Stockholm, Sweden
Dec 2024	**Around Local Index Theory** Intersection bundles and the Cappell–Miller torsion for Riemann surfaces. Mini-course on joint work with Gerard Freixas i Montplet.	Paris, France
Nov 2024	**Algebra and Friends** Milnor Number Inequalities in Hypersurface Singularities	Borås, Sweden
May 2024	**Complex Geometry Workshop** On some singularity invariants from analytic torsion	Aarhus, Denmark
Sep 2021	**Arakelov geometry intercity conference** Genus one mirror symmetry and the arithmetic Riemann–Roch theorem	Regensburg, Germany
2019	**Workshop Alkage** On an invariant for Calabi–Yau manifolds through analytic torsion	Grenoble

Complete List of Publications

2026 **Spectrum of singularities, analytic torsion and Calabi–Yau degenerations**
Preprint · With Gerard Freixas i Montplet.

2026 **Base change conductors through intersection theory and quotient singularities**
Documenta Mathematica · With Lars-Halvard Halle, Johannes Nicaise.

2025 **The spectral genus of an isolated hypersurface singularity and its relation to the Milnor number and analytic torsion**
Documenta Mathematica · With Gerard Freixas i Montplet.

2025 **The Deligne-Riemann-Roch isomorphism**
Preprint · With Gerard Freixas i Montplet.

2024 **Erratum to: Discriminants and Artin conductors (J. Reine Angew. Math. 712 (2016), 107–121)**
J. Reine Angew. Math. 814 (2024), 283–284 (Erratum (MR4793346))

2024 **Deligne-Riemann-Roch and intersection bundles**
J. Éc. polytech. Math. 11 (2024), 247–361 · With Gerard Freixas i Montplet.

2024 **Genus one mirror symmetry for intersection of two cubics in $\mathbb{P}^5$**
Preprint · With Mykola Pochekai.

2022 **Complex Chern–Simons bundles in the relative setting**
Astérisque (Accepted for publication) · With Gerard Freixas i Montplet, Richard Wentworth.

2022 **On genus one mirror symmetry in higher dimensions and the BCOV conjectures**
Forum Math. Pi 10(e19) (2022), 1–53 · With Gerard Freixas i Montplet, Christophe Mourougane.

2021 **Global representation of Segre numbers by Monge–Ampère products**
Mathematische Annalen 380(1-2) (2021), 349–391 · With Mats Andersson, Håkan Samuelsson Kalm, Elizabeth Wulcan, Alain Yger.

2021 **Spaces of norms, determinant of cohomology and Fekete points in non-Archimedean geometry**
Advances in Mathematics 378 (2021), 107501–107501 (Article ID 107501) · With Sébastien Boucksom.

2021 **Nonproper intersection products and generalized cycles**
Eur. J. Math. 7(4) (2021), 1337–1381 · With Mats Andersson, Håkan Samuelsson Kalm, Elizabeth Wulcan, Alain Yger.

2021 **BCOV invariants of Calabi–Yau manifolds and degenerations of Hodge structures**
Duke Mathematical Journal 170(3) (2021), 379–454 · With Gerard Freixas i Montplet, Christophe Mourougane.

2020 **Kähler quantisation of vortex moduli**
Letters in Mathematical Physics 110(3) (2020), 659–693 · With Nuno Romão.

2018 **Singularities of metrics on Hodge bundles and their topological invariants**
Algebraic Geometry 5(6) (2018), 742–775 · With Gerard Freixas i Montplet, Christophe Mourougane.

2016 **Discriminants and Artin conductors**
Journal für die reine und angewandte Mathematik (Crelle) 712 (2016), 107–121

2015 **A logarithmic interpretation of Edixhoven's jumps for Jacobians**
Advances in Mathematics 279 (2015), 532–574 · With Lars-Halvard Halle, Johannes Nicaise.

2013 **Degenerating Riemann surfaces and the Quillen metric**
International Mathematics Research Notices 2013(2) (2013), 347–361

2009 **Un isomorphisme de type Deligne–Riemann–Roch**
Comptes Rendus Mathematique 347(19-20) (2009), 1115–1118

2008 **On the Brauer–Manin obstruction for zero-cycles on curves**
Acta Arithmetica 135(2) (2008), 99–110 · With Victor Scharaschkin.